DR-2 cross-scale induction v1.1 – Besicovitch-mean repair of the

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

dr2-decoupling-closure v1.1 proved DR-2 for δ-separated $Q \subset S^2$ (T6 conditional on ℓ^2-decoupling) and left the arbitrary-Q (non-separated) case at T4, naming the cross-scale energy summation as the residual. v1.0 of THIS note attempted that step but its iteration lemma re-used the identity $\|f_\theta\|_{L^4}^4 = E_+(Q_\theta)$, which is FALSE for $q \in S^2$ (the frequencies are not in $\mathbb{Z}^3$, so $\int_{[0,1]^3} |f|^4$ is a product of sinc factors, not the additive-energy Kronecker count). v1.1 repairs the bridge with the Besicovitch mean and re-grades honestly. It remains conditional on the same decoupling black box and reproves nothing about decoupling itself.

2. The Besicovitch-mean bridge and the iteration inequality

2.1 The exact additive-energy identity (no integrality)

For a finite set $Q \subset \mathbb{R}^3$ put $f_Q(x) = \sum_{q \in Q} e(q \cdot x)$, $e(t) = e^{2\pi i t}$, and define the Besicovitch mean

$$M(F) := \limsup_{R \to \infty} \frac{1}{|B_R|} \int_{B_R} F(x)\, dx.$$

Since $M\big(e(v \cdot x)\big) = \mathbf{1}[v = 0]$ for every $v \in \mathbb{R}^3$ (for $v \neq 0$ the oscillatory integral is $O(R^2)$ against the volume $\sim R^3$, hence $\to 0$; for $v = 0$ it is 1), expanding $|f_Q|^4 = \sum_{i,j,k,l} e\big((q_i + q_j - q_k - q_l) \cdot x\big)$ gives

$$\boxed{M\big(|f_Q|^4\big) = \#\{(i,j,k,l) : q_i + q_j = q_k + q_l\} = E_+(Q)}$$

EXACTLY, for ANY finite Q – integer or not. This is the identity v1.0 was missing: the additive energy is the Besicovitch mean of $|f_Q|^4$, not a torus L^4 norm. The same holds cap-wise, $M(|f_\theta|^4) = E_+(Q_\theta)$.

2.2 The iteration lemma

Lemma (Besicovitch–decoupling iteration). Let $Q \subset S^2$ be finite, $\delta > 0$, and let $\{\theta\}$ be the partition of S^2 into $\delta^{1/2}$-caps, $Q_\theta = Q \cap \theta$. Then, conditional on ℓ^2-decoupling at the critical exponent $p = 4$,

$$E_+(Q) \ \lesssim_\epsilon \ \delta^{-\epsilon} \Big(\sum_\theta E_+(Q_\theta)^{1/2} \Big)^2.$$

Derivation. Bourgain–Demeter ℓ^2-decoupling is a LOCAL (ball) statement: on any ball B_R with $R \sim \delta^{-1}$ and a Schwartz weight $w_{B_R} \geq 0$ adapted to B_R,

$$\|f_Q\|_{L^4(w_{B_R})} \lesssim_\epsilon \delta^{-\epsilon} \Big(\sum_\theta \|f_\theta\|_{L^4(w_{B_R})}^2 \Big)^{1/2}.$$

Raise to the fourth power and tile $\mathbb{R}^3$ by translates $\{B_R + y\}$ of B_R; each translate obeys the same inequality with the same constant. Averaging the fourth powers over the translates and exhausting $\mathbb{R}^3$ ($R \to \infty$ through the tiling) sends the local averages to the Besicovitch means: the left side to $M(|f_Q|^4) = E_+(Q)$ and each cap term to $M(|f_\theta|^4) = E_+(Q_\theta)$. Hence $E_+(Q) \lesssim_\epsilon \delta^{-\epsilon}(\sum_\theta E_+(Q_\theta)^{1/2})^2$. □

Cited (not reproduced) step R1. The decoupling-compatible passage from the ball inequality to the Besicovitch-mean inequality – i.e. that translate-averaging the LOCAL cap norms $\|f_\theta\|_{L^4(w_{B_R})}$ yields the GLOBAL $E_+(Q_\theta)$ with a uniform constant – is the standard local-to-global argument for decoupling and is invoked here, not reproved. This is the honest replacement for v1.0's false one-line identity: the bridge is now correct in form (Besicovitch mean) but its decoupling-compatible execution is cited.

Numerical consistency (dr2_decoupling_iteration.py v1.1, 4/4, ILLUSTRATIVE). A code-audit gate first verifies the additive-energy estimator is EXACT (a random sphere set has exactly $2N^2 - N$ trivial quadruples). On a PROXY cap partition (square (x, y) bins, NOT geodesic $\delta^{1/2}$-caps) the constant $K = E_+(Q)/(\sum_\theta E_+(Q_\theta)^{1/2})^2$ stays ≤ 1.94 and scale-stable (within $\times 1.25$ across $n = 2, 4, 8$). This is consistency with a mild $\delta^{-\epsilon}$ constant, NOT a faithful decoupling test and NOT a proof; no tier rests on it.

3. The induction-on-scales

Base case (separated). If Q is $\delta^{1/2}$-separated every cap holds ≤ 1 point, $E_+(Q_\theta) \leq 1$, so the Lemma gives $E_+(Q) \lesssim_\epsilon \delta^{-\epsilon} N^2$; with $\delta \geq N^{-O(1)}$, $E_+(Q) \lesssim_\epsilon N^{2+\epsilon}$ (this is v1.1 §3).

Recursion (clustered). If a cap θ holds $m_\theta > 1$ points, $E_+(Q_\theta)$ is a sub-problem on a $\delta^{1/2}$-cap. The additive energy is affine-invariant (R-023): for the affine map T_θ normalising the cap, $E_+(Q_\theta) = E_+(T_\theta Q_\theta)$ EXACTLY. The crucial honesty point (v1.0 overstated this): $T_\theta \theta$ is a *uniformly-curved* C^2 *graph patch*, NOT literally the standard unit paraboloid. What survives is (i) the energy is preserved exactly by the affine map, and (ii) ℓ^2-decoupling is STABLE under nondegenerate C^2 perturbations of the paraboloid, so the decoupling constant for the rescaled patch is uniform. Thus the Lemma applies to the rescaled sub-instance with $m_\theta < N$ points, and the recursion proceeds. It terminates when every active cap is separated, depth $O(\log N)$ for sets with $\geq N^{-O(1)}$ separations.

4. The two residuals and status

The cross-scale step is now WRITTEN with a correct bridge, and the grade rests on exactly TWO cited-not-reproduced ingredients:

Residual R1 (local-to-global). The decoupling-compatible translate-averaging that turns the ball inequality into the Besicovitch-mean inequality (§2.2), i.e. the local cap norm $\to E_+(Q_\theta)$ bridge with a uniform constant. Standard for decoupling; cited.

Residual R2 (multi-scale bookkeeping). Summing the inductive bounds $E_+(Q_\theta) \lesssim_\epsilon m_\theta^{2+\epsilon}$ over $O(\log N)$ levels without the per-level $\delta^{-\epsilon}$ accumulating to $N^{O(1)}$. The standard Bourgain–Demeter / Bourgain–Guth organisation (geometric per-level ϵ_k, or the fact that ℓ^2-decoupling is already the multi-scale theorem carrying a single $\delta^{-\epsilon}$ at the critical scale) keeps the total at $N^{O(\epsilon)}$. Cited, not reproduced.

Status. With R1 and R2 both cited, the unrestricted DR-2 is a STRUCTURALLY-COMPLETE reduction whose two remaining ingredients are standard decoupling machinery. Honest tier: **T4+ STRONG EVIDENCE**, and an explicit **T5-candidate** pending operator acceptance of R1+R2 as textbook-standard. v1.0's outright T5 is **WITHDRAWN** (it rested on a false bridge). Whether accepting R1+R2 warrants T5 (structural) or, together with the separated base, T6 PROVED CONDITIONAL on decoupling and a DR2-SHARE flip, is an OPERATOR decision. No flip is made here: DR2-SHARE stays OPEN; B5 T5; B1 T6; H-ADM-COH retained.

5. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3, code-discipline 6)

Three concrete objections, found before submission:

(α) *"The Besicovitch mean fixes the GLOBAL identity, but the decoupling RHS carries LOCAL cap norms $\|f_\theta\|_{L^4(w_{B_R})}$; you still silently identify those with $E_+(Q_\theta)$."* VALID and now explicitly RECORDED as Residual R1 (§2.2, §4). The repair over v1.0 is honesty, not elimination: v1.0 asserted a false EXACT identity in one line; v1.1 routes the identity through the Besicovitch mean (correct) and flags the decoupling-compatible local-to-global passage as the cited step. The grade (T4+, not T5/T6) reflects that R1 is cited.

(β) *"Decoupling is stated on the paraboloid; S^2 and the rescaled curved patches are not the paraboloid."* DISMISSED as an obstruction, RECORDED as a hypothesis: Bourgain–Demeter decoupling holds for any compact hypersurface with nonvanishing Gaussian curvature, and is stable under nondegenerate C^2 perturbations; S^2 qualifies and the affine-normalised caps are uniformly-curved C^2 graphs. §3 now states this rather than the v1.0 overclaim "unit paraboloid patch". No additional residual is incurred beyond the decoupling black box already assumed.

(γ) *"The per-level $\delta^{-\epsilon}$ could accumulate to $N^{O(1)}$ over $O(\log N)$ levels, destroying $N^{2+\epsilon}$."* This is the SHARP objection and is exactly Residual R2. It is not dismissed: it is the reason the grade is T4+/T5-candidate and not T6. The numerics (K scale-stable within $\times 1.25$) are consistent with a non-accumulating constant but do NOT prove it; R2 is the cited Bourgain–Demeter/Bourgain–Guth bookkeeping.

(δ) *"The numerics could be mistaken for evidence of closure."* GUARDED: the script v1.1 carries an exact-E_+ code-audit gate and an explicit PROXY-partition caveat (square bins are not geodesic caps); the note labels the numerics ILLUSTRATIVE throughout and rests no tier on them.

Quantitative sanity check (mandatory). Magnitude: $K \leq 1.94$ (mild constant) and $\sum_\theta E_+(Q_\theta) \approx 2 \times 10^2 \ll E_+(Q) \approx 10^4$ (the iteration genuinely reduces structure). Sign/direction: upper bound, all terms positive. Limit-case: separated Q recovers the v1.1 base case exactly. Estimator exactness: random $E_+ = 2N^2 - N$ verified for $N = 16, 32, 64$ (no binning artefact). Consistency: K stable across cap scales matches a mild $\delta^{-\epsilon}$.

6. Result footer

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Result ID:              B5 / DR-2 cross-scale induction v1.1
                        (Besicovitch-mean repair)
Precise statement:      For finite Q in S^2, M(|f_Q|^4)=E_+(Q) exactly
                        (Besicovitch mean; no integrality). Conditional on
                        l2-decoupling at p=4, the iteration
                        E_+(Q) <=_eps delta^{-eps}(sum_theta sqrt E_+(Q_theta))^2
                        follows by ball-decoupling + translate-averaging to
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	the mean; with the R-023 affine rescaling (caps are uniformly-curved C^2 graph patches handled by decoupling stability, NOT the literal paraboloid) it reduces arbitrary finite Q to the separated base (-> N^{2+eps}).
Two cited residuals:	R1 local-to-global translate-averaging (local cap norm -> E_+(Q_theta), uniform constant); R2 multi-scale eps-bookkeeping over O(log N) levels -> total N^{O(eps)}.
Scope:	Unrestricted-class sphere additive energy; cross-scale step of DR-2 / DR2-SHARE.
Dependencies:	l2-decoupling (BLACK BOX) + its standard stability and local-to-global machinery (R1) + multi-scale bookkeeping (R2); R-023 (affine-invariance).
Evidence grade:	DERIVED with a CORRECT bridge (Besicovitch mean), conditional on decoupling and on two cited standard ingredients (R1,R2); + ILLUSTRATIVE numerics (dr2_decoupling_iteration.py v1.1 4/4, proxy partition).
Reproduction command:	python codes/vacuum/dr2_decoupling_iteration.py
Expected output:	4/4 PASS (incl. exact-E_+ audit gate); K <= 1.94 scale-stable on the PROXY partition (illustrative).
Falsification gate:	A finite Q with K growing polynomially in N (would contradict the decoupling exponent); not observed.
Tier before / after:	DR-2 unrestricted: v1.0 T5 WITHDRAWN -> T4+ STRONG EVIDENCE (T5-candidate pending operator acceptance of R1+R2). NO flip. Separated case stays T6 conditional. B5 T5, B1 T6, H-ADM-COH retained, DR2-SHARE OPEN.
No-overclaim statement:	NOT a DR-2 closure, NOT T5-accepted, NOT T6. The bridge is now correct (Besicovitch mean) but two standard ingredients (R1,R2) are cited, not reproduced. Decoupling is invoked, not reproved. No gate/tier/ hypothesis action; operator authorises any flip.
Next required action:	Operator decision on whether R1+R2 are accepted as textbook-standard (-> T5 or T6 PROVED CONDITIONAL and a possible DR2-SHARE flip), or reproduce R1+R2 in full.